

2023



Reducing Developer Overhead by 50% by Migrating to Amazon EKS with Mobileye

Learn how Mobileye, a leader in autonomous-driving and driver-assist technologies, transformed its backend into a flexible, dynamic stack using Amazon EKS.

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50%

reduction in developer overhead, freeing engineers to innovate

Minimizes

time spent debugging



2

Supports

open-source solutions

Scalable

Provides scalable, flexible resource use

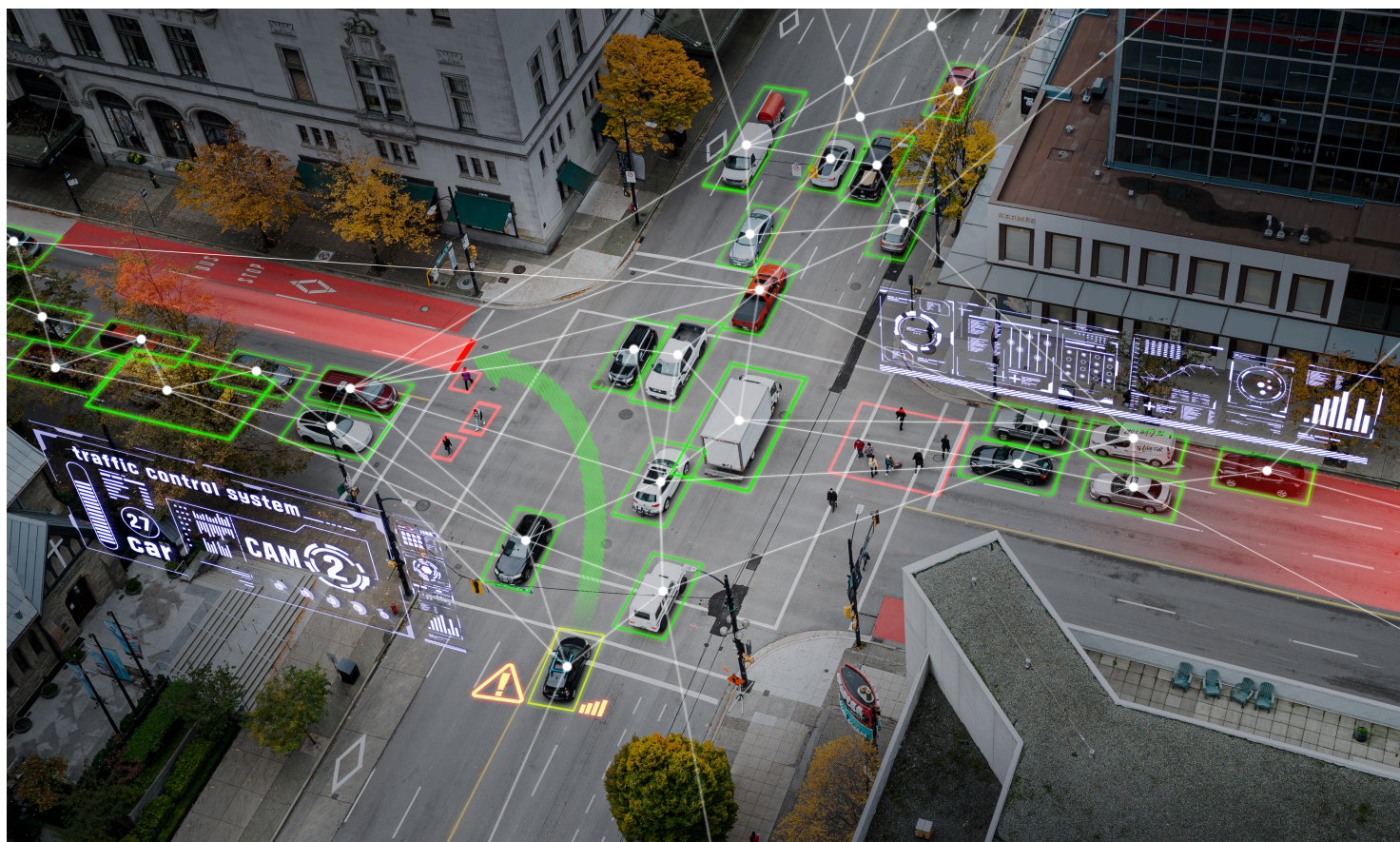
400k

virtual CPUs run on thousands of instances concurrently with two clusters

Overview

Autonomous-driving technology company [Mobileye Global Inc.](#) (Mobileye) was looking to empower its Road Experience Management (REM) team with a powerful compute solution that could scale to the demand of its high-definition mapping while providing its engineering teams with configuration flexibility. The team's goal was to build a scalable stack that would support the company's time to market by delivering the advanced capabilities of Kubernetes, including resource isolation and dynamic configuration.

Having built its environments on Amazon Web Services (AWS), the REM team migrated its workloads to [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed Kubernetes service to run Kubernetes in the AWS Cloud and on-premises data centers. Mobileye worked alongside the AWS team to transform its backend into a flexible, dynamic stack that helps its teams debug software updates and innovate faster than before.



Mobileye leads the mobility revolution with its autonomous-driving and driver-assist technologies, harnessing world-renowned expertise in computer vision, artificial intelligence, mapping, and data analysis. As of October 2023, more than 150 million vehicles worldwide have been built with Mobileye technology inside. Mobileye's REM team focuses on creating high-definition road maps that autonomous-driving vehicles use for navigation.

The REM team built its infrastructure using a variety of open-source tools, such as Apache Spark. However, its previous software stack provided little flexibility or configurability. "We wanted the ability to make changes on the fly without redeploying or reconfiguring compute resources," says Ben-Tzion Lifshitz, software developer at Mobileye. "We couldn't configure different Apache Spark drivers or diverse compute sizes in a single cluster." As a result, changes that engineers made to the configuration of one application could impact other applications in production. Mobileye's engineering teams will benefit from each application running in an isolated environment. "It will dramatically reduce our debugging times," says Ofer Eliassaf, senior director of REM at Mobileye. "The approach helps us avoid situations where one Spark application interferes with another, freeing up our engineers to solve complex problems."

Mobileye decided to migrate its workloads to Apache Spark on Kubernetes using Amazon EKS. "Using Amazon EKS, we removed the additional burden from our DevOps teams," says Eliassaf. "Kubernetes is very complex software, and using Amazon EKS makes it simpler to install and work with Kubernetes than I had imagined possible."

Solution | Rearchitecting Mobileye's Tech Stack on AWS for Configurability and Observability

In February 2023, Mobileye began migrating to Amazon EKS, testing the performance of small-scale workloads. For its machine learning and data workloads, the REM team shortened its development time by using [Data on EKS](#) (DoEKS), an open-source project aimed at streamlining and accelerating the process of building, deploying, and scaling data workloads on Amazon EKS. Mobileye also engaged the AWS team for technical support with scaling. "With the support of the AWS team, we adapted configurations that worked best for our existing environment," says Lifshitz.

Mobileye also used [DoEKS Blueprints](#), which incorporate guidance and tools to support the changing requirements of data-intensive workloads on Kubernetes. The REM team used DoEKS Blueprints on Kubernetes to deploy Apache Spark solution to run with the Spark operator.



The team configured Amazon EKS to run alongside Karpenter. By using Karpenter, Mobileye has unlocked automatic scaling capabilities and can diversify its instance types in [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for virtually any workload. For example, the company runs the majority of its workloads using [Amazon EC2 Spot Instances](#), which businesses can use to take advantage of unused Amazon EC2 capacity in the AWS Cloud at up to a 90 percent discount compared with Amazon EC2 On-Demand prices. Using Amazon EKS, Mobileye scaled to run more than 400,000 virtual CPUs concurrently on thousands of Amazon EC2 instances across two clusters. “We can automatically scale much better using more diverse instances,” says Lifshitz. “We can also run less critical jobs on Spot Instances and diminish instances in 10 minutes after we’re finished using them.”

With greater flexibility to configure containers, the REM team can also boost productivity. “Using Amazon EKS, we can change the way our developers work,” says Eliassaf. “We can provide them with a better experience, which will improve their performance and our time to market. They can more easily use open-source tools and AWS services.”

Outcome | Reducing Developer Overhead by 50% with Managed Services

The Mobileye REM team rolled out its new architecture to production in November 2023. In addition to increased visibility, the team has benefited from more reliable infrastructure performance as a result of deploying Amazon EKS across multiple AWS Regions.

Further, the REM team estimates that it has cut its developer overhead in half by choosing to collaborate with the AWS team on this project. “If we’d had to complete a project like this on our own, we would’ve had to double our DevOps team,” says Eliassaf. “Using Amazon EKS, we can deploy Kubernetes elegantly and cost effectively with fewer engineers.”

About Mobileye Global Inc.

Mobileye leads the mobility revolution with its autonomous-driving and driver-assist technologies, offering world-renowned expertise in computer vision, artificial intelligence, mapping, and data analysis.

AWS Services Used

Amazon EKS

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

[Learn more »](#)



Amazon EC2 Spot Instances

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